

Two satellites — Solution

X22 (2022) — English translation

A1

0.40 pts

What is the speed of the satellites v at the moment of launch? Express your answer in terms of G , M_E and R_E .

Since when only one of the satellites is launched into orbit, its trajectory will be a parabola, then the speed v is equal to the second escape velocity for the Earth, that is:

Answer:

$$v = \sqrt{\frac{2GM_E}{R_E}}$$

A2

0.60 pts

At what distance from the center of the Earth R_0 is the point P of intersection of the satellite trajectories? Express your answer using R_E .

To move along the parabola

$$r = \frac{p}{1 + \cos \theta}$$

in point launch satellites

$$R_E = \frac{p}{2}$$

from which

$$p = 2R_E$$

where p - focal parameter. Further, since orbit symmetric, parameter θ equal $\frac{\pi}{2}$ for both satellites, from where

Answer:

$$R_0 = 2R_E$$

A3

0.80 pts

What is the angle φ between the directions of satellite velocities at point P ?

For each of the satellites at point P can be found from the law of conservation of energy

$$v_P = \frac{v}{\sqrt{2}}$$

From angular momentum conservation

$$vR_E = v_{\perp}R_0 = 2Rv_{\perp}$$

From which

$$\frac{v_{\perp}}{v_P} = \frac{1}{\sqrt{2}}$$

Means, vector velocity in point P constitute angle $\alpha = \frac{\pi}{4}$ with radius vector. Then from symmetry orbit

$$\varphi = 2\alpha = \frac{\pi}{2}$$

Also, the answer can be easily obtained from the optical property of a parabola.

Answer:

$$\varphi = 2\alpha = \frac{\pi}{2}$$

A4

1.50 pts

Find the time t_0 after which the satellites reach point P . Express your answer using G , R_E and M_E .

The expression for the sectorial speed is as follows

$$\frac{dS}{dt} = \frac{L}{2m} = \sqrt{\frac{GM_E R_E}{2}}$$

From which

$$t_0 = \frac{S}{\sqrt{\frac{GM_E R_E}{2}}}$$

where S - area, swept radius-vector from point launch to point P . Direct axis y by direction on point launch from center Earth, and axis x perpendicularly it. From the equation parabola in polar coordinate

$$p = r(1 + \cos \theta) = r + y$$

from which

$$r^2 = y^2 + x^2 = p^2 + y^2 - 2py$$

Obtain

$$y(x) = \frac{p}{2} - \frac{x^2}{2p} = R - \frac{x^2}{4R}$$

Then

$$S = \int_0^{2R_E} y dx = \int_0^{2R} \left(R - \frac{x^2}{4R} \right) dx = 2R^2 - \frac{8R^3}{12R^2} = \frac{4R^2}{3}$$

from here

Answer:

$$t_0 = \frac{4\sqrt{2}}{3} \sqrt{\frac{R_E^3}{GM_E}}$$

B1

0.70 pts

Assuming that near the point P the satellites moved along straight lines, the directions of which coincided with the directions of their velocities at this point, find the relative speed of the satellites v_0 , as well as the minimum distance b between them, neglecting their gravitational interaction. Express your answer using v and τ .

Let us denote by v_P the speed of the satellites at point P . The speeds of the satellites at point P are mutually perpendicular, which means the relative speed is $v_0 = v_P \sqrt{2}$ and

At the moment when the satellite 1 passes the point P , the satellite 2 is at a distance $l = v_P \tau$ from it, and the angle between the relative velocity vector and the relative radius vector of the particle 1 is equal to $\frac{3\pi}{4}$. Thus $b = \frac{l}{\sqrt{2}}$ and

Answer:

$$v_0 = v$$

B2

1.00 pts

Find u . Express your answer using v .

The absolute values of the initial and final relative velocities of the satellite 1 coincide. From the law of addition of velocities

$$\vec{u} = \vec{v}_{2P} + \vec{v}_{\text{rel}}$$

Let β - angle between vector $\vec{u}$ and $\vec{v}_{\text{rel}}$. Then from theorem sine

$$\frac{v_P}{\sin(\beta)} = \frac{v_0}{\sin\left(\frac{\pi}{4}\right)}$$

from which

$$\beta = \frac{\pi}{6}$$

For v_1 obtain

$$u = v_P \cos \frac{\pi}{4} + v_0 \cos \beta$$

Answer:

$$u = v_P \cos \frac{\pi}{4} + v_0 \cos \beta = v \cdot \frac{1 + \sqrt{3}}{2}$$

B3

2.30 pts

Find τ . Express your answer in terms of G , M_2 , M_3 and R_E .

For eccentricity we have

$$e = \frac{1}{\cos\left(\frac{\beta}{2}\right)} = \frac{1}{\cos\left(\frac{\pi}{12}\right)}$$

Expression for eccentricity

$$e = \sqrt{1 + \frac{v^4 b^2}{G^2 M_2^2}}$$

From which

$$\frac{4}{2 + \sqrt{3}} = 1 + \frac{v^4 b^2}{G^2 M_2^2}$$

Or

$$2 - \sqrt{3} = \frac{v^2 b}{GM_2} = \frac{2bM_E}{R_E M_2}$$

From

Answer:

$$\tau = (2 - \sqrt{3}) \sqrt{\frac{R_E^3}{2GM_E}} \cdot \frac{M_2}{M_E}$$

C1

1.00 pts

Find the speed v_∞ of satellite 1 at an infinite distance from the Earth. Express your answer using v .

From the law of conservation of energy

$$\frac{E}{M_1} = \frac{u^2}{2} - \frac{GM_E}{2R_E} = \frac{v_\infty^2}{2}$$

Account for obtain earlier relation

$$v^2 \cdot \frac{(1 + \sqrt{3})^2}{8} - \frac{v^2}{4} = \frac{v_\infty^2}{2} = v^2 \cdot \frac{1 + \sqrt{3}}{4}$$

From

Answer:

$$v_\infty = v \sqrt{\frac{1 + \sqrt{3}}{2}}$$

C2

1.50 pts

Find the angle φ_∞ between vectors $\vec{u}$ and $\vec{v}_\infty$.

The expression for the orbital eccentricity is as follows:

$$e = \sqrt{1 + \left(\frac{v_{\infty} \cdot u \cdot 2R}{GM_E} \right)^2} = \sqrt{1 + \left((1 + \sqrt{3})^{\frac{3}{2}} \cdot \sqrt{2} \right)^2}$$

From

Answer:

$$\varphi_{\infty} = \frac{\pi}{2} - \arccos \left(\frac{1}{\sqrt{1 + 2(1 + \sqrt{3})^3}} \right)$$

C3

0.20 pts

Is it really possible to put a satellite into orbit this way? Justify your answer.

Gravity is the “weakest” of the four fundamental interactions. The main manifestation of gravity that we encounter in everyday life is the Earth’s gravitational field; Although any objects with mass interact with each other in this way, this interaction can almost always be neglected. Let us give quantitative estimates of the question of whether two artificial Earth satellites can so significantly influence each other’s trajectories. The mass of the heaviest *capable move* structures built by mankind can reach several hundred thousand tons (for example, aircraft carriers). Let’s assume that the mass is $M_2 \sim 10^9 \text{kg}$. Then, to perform the maneuver specified in the problem, the impact parameter b in order of magnitude can be estimated as

$$b \sim \frac{M_2}{M_3} R_3$$

Use $M_3 = 6 \cdot 10^{24} \text{kg}$, $R_3 = 6.4 \cdot 10^6 \text{ m}$, obtain:

$$b \sim 10^{-9} \text{m},$$

which is comparable to interatomic distances. As one would expect, in reality the gravitational attraction of the satellite is negligible.

Answer: No